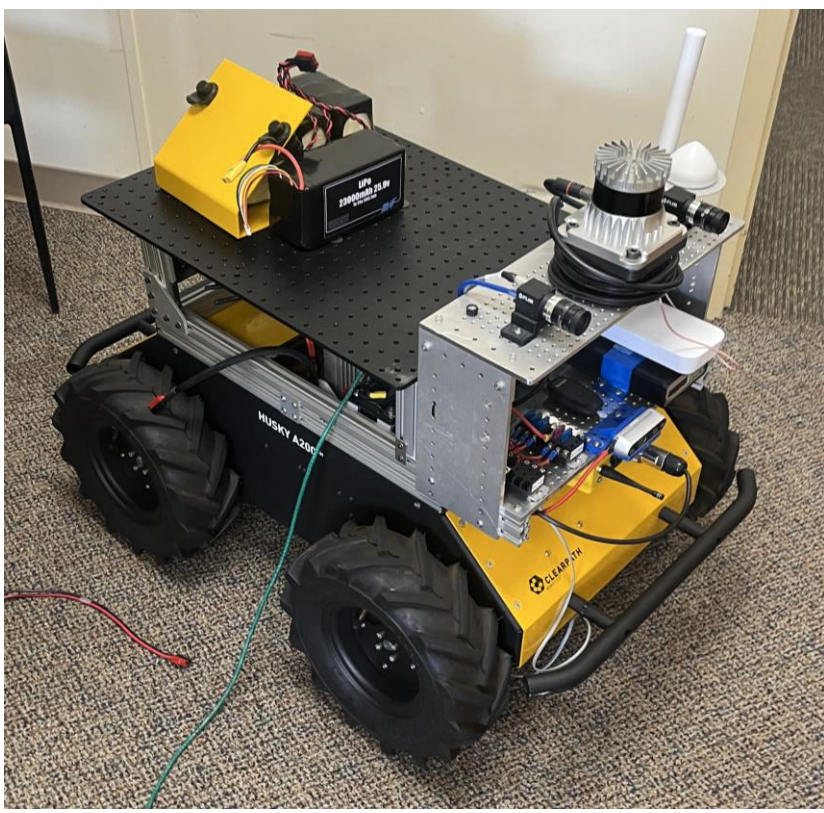




CYBER-PHYSICAL TEAM: SYSTEM CALL PATTERN RECOGNITION

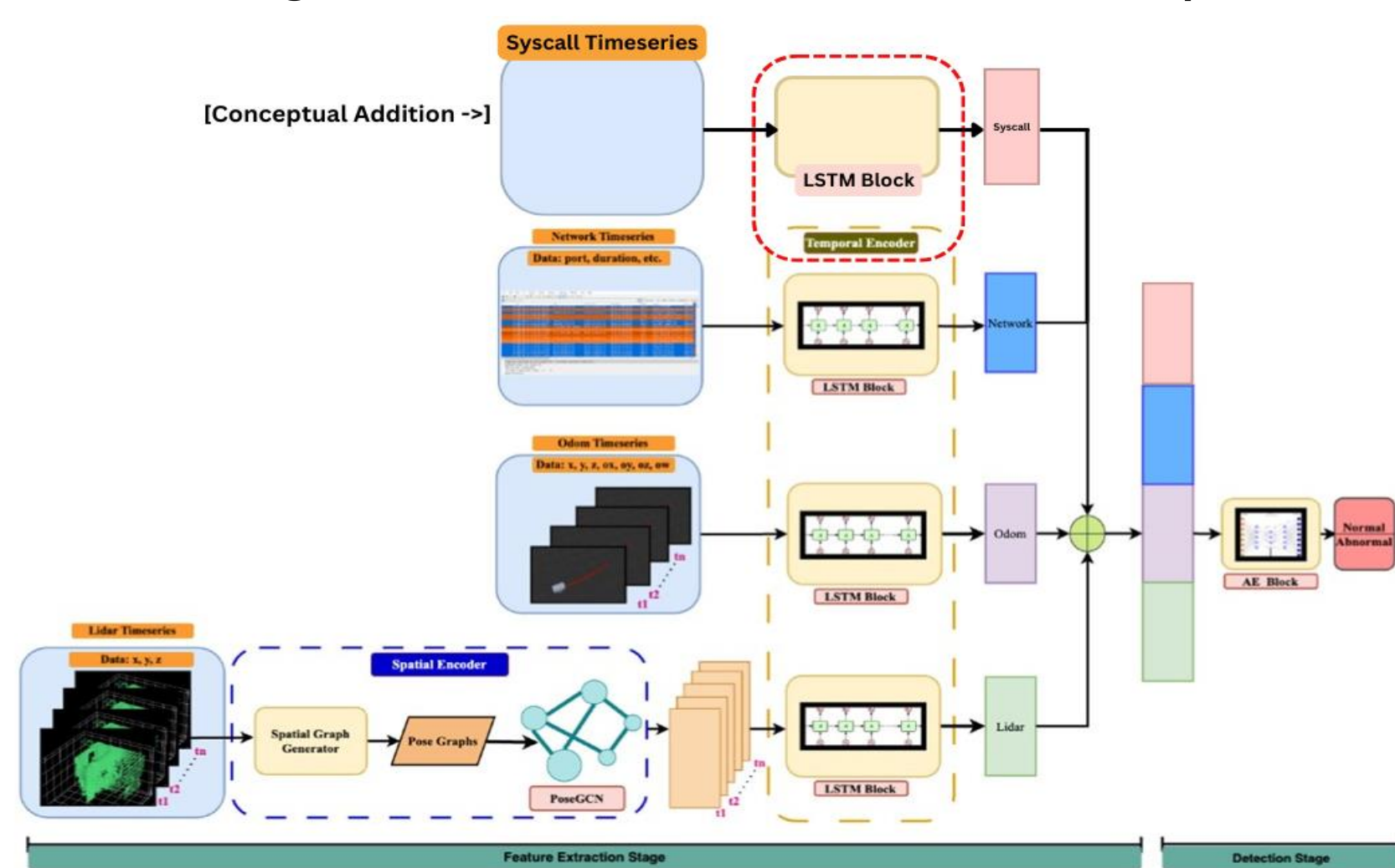
Project Background

University of Maryland (UMD) designed an unsupervised machine learning model, Anomaly Detection using Spatio-Temporal Robot Data (U-ASTROD), to extract features and detect irregularities in three current modalities' data – and network – from a Husky unmanned ground vehicle (UGV). The findings are reported back to an interactive dashboard that monitors the robot's status, providing security and resilience to the Husky UGV.



Approach

- Integrate system call data handling into existing U-ASTROD model
 - Find and extract patterns & features from UMDHusky's historical system call data using an Long-Short Term Memory (LSTM) model
 - Train, validate, and test the model on this historical data
 - Move onto real-time system call data integration
- Use an n-gram as a base model for comparison



Results

- Wrote a script that captured .bag files, a file type created by the Robot Operating System (ROS) that the UMDHusky runs, and converted them into CSVs.
- Wrote a module that split the original CSV into multiple categories for better model comprehension. Standardized Unix epoch time, split timestamp, and padded PID.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Time	data												
2	1719506492.2940538	point_cloud_pipeline/point_cloud_pipeline::[pid 4395] 12:41:32.178776 futex(0x55895d1556e8, FUTEX_WAKE_PRIVATE, 1) = 1 <0.000086>												
3	1719506492.2940576	point_cloud_pipeline/point_cloud_pipeline::[pid 4395] 12:41:32.178904 futex(0x55895c0c0f68, FUTEX_WAIT_BITSET_PRIVATE, 0, {tv_sec=222:												
4	1719506492.2941835	goto_object_behavior::[pid 4363] 12:41:32.180267 <... clock_nanosleep resumed>0x7ffe08a00b30) = 0 <0.050244>												
5	1719506492.294187	goto_object_behavior::[pid 4363] 12:41:32.180424 clock_nanosleep(CLOCK_REALTIME, 0, {tv_sec=0, tv_nsec=500000000}, <unfinished ...>												

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Epoch Time	Hours	Minutes	Seconds	Milliseconds	Node	PID	Syscall Data							
2	-1.548107579	12	41	32	178	point_cloud_pipeline/point_cloud_pipeline	04395	futux(0x55895d1556e8, FUTEX_WAKE_PRIVATE, 1) = 1 <0.000086>							
3	-1.548106991	12	41	32	178	point_cloud_pipeline/point_cloud_pipeline	04395	futux(0x55895c0c0f68, FUTEX_WAIT_BITSET_PRIVATE, 0, {tv_sec=2221, tv_nsec=36626							
4	-1.548087607	12	41	32	180	goto_object_behavior	04363	<... clock_nanosleep resumed>0x7ffe08a00b30) = 0 <0.050244>							
5	-1.548087056	12	41	32	180	goto_object_behavior	04363	clock_nanosleep(CLOCK_REALTIME, 0, {tv_sec=0, tv_nsec=500000000}, <unfinished ...>							

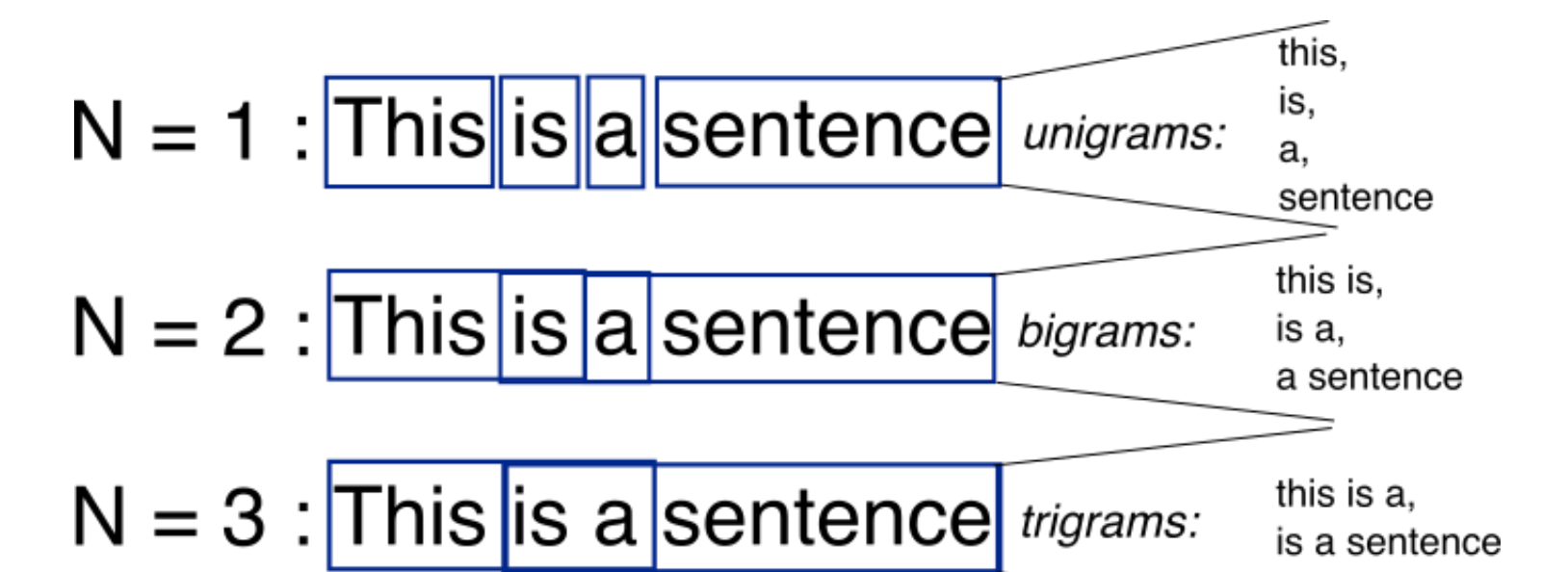
- Tokenized 'Node' and 'Syscall Data' categories due to their non-numerical data, which Long-Short Term Memory models cannot handle.

Sample tokenized sequences for 'Syscall Data': [[8, 159, 8, 15, 12, 5, 5, 1, 528], [8, 489, 8, 50, 34, 12, 1, 2, 4, 280, 2, 6, 3442, 8, 34, 51, 52, 19], [3, 67, 16, 88, 1, 1, 3443], [3, 67, 3, 92, 1, 2, 4, 1, 2, 6, 138, 19], [8, 30, 8, 15, 12, 5, 5, 1, 23]]
 Sample tokenized sequences for 'Node' Data: [[9, 6, 10, 9, 6, 10], [9, 6, 10, 9, 6, 10], [18, 19, 20], [18, 19, 20], [4, 1, 1, 5]]

- Utilized GloVe's embedding vectors to encode the 'Node' and 'Syscall Data' categories that featured non-numerical text.

Future Work

- The model still needs to be adapted to work with live data coming from the UMDHusky
 - Would allow real-time feature extraction which in turn, would assist with real-time anomaly detection
 - Promotes better resilience to potential threats and attacks
- Observe the accuracy differences between GloVe and Word2Vec's embedding vectors
 - Attempt to concatenate both embedding vectors to see the effect on accuracy
- N-gram has not yet been created. This is necessary to establish a baseline for proving the usefulness of the LSTM model.



Acknowledgments

I would like to give special thanks to my project mentor, Dr. Rob Erbacher, and the University of Maryland's Center for Advanced Transportation Technology (CATT) lab members, Mahshid Noorani and Jack Mirenzi, whose direction and guidance steered my research. Additionally, I thank the CATT Lab for hosting my team and I there, and for providing us access to the UMDHusky as needed.



FOR FURTHER INFORMATION:

U.S. ARMY COMBAT CAPABILITIES
DEVELOPMENT COMMAND
ARMY RESEARCH LABORATORY:
ARL.ARMY.MIL

POINT OF CONTACT:
Gabriela Schweers
gschwee1@umbc.edu
443-299-8761





FOR FURTHER INFORMATION:

U.S. ARMY COMBAT CAPABILITIES
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